Vincent T. Cook

HVAC ENGINEERING INTERNSHIP

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Aerospace engineering senior seeking an HVAC engineering internship. Background includes airflow testing, mechanical CAD and schematics, power-efficiency studies, controls coursework, and electrical troubleshooting.

EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

TECHNICAL SKILLS

Design: SolidWorks, Autodesk Inventor, 3D CAD, schematics, assembly instructions, stress analysis
Analysis: MATLAB/Simulink, Python, airflow measurements, CFD, thermal testing, circuit troubleshooting

ENGINEERING EXPERIENCE & PROJECTS

SAE Spartan Racing | EV Powertrain Intern | Aug 2023-May 2025
- Created CAD and assembly instructions for an electric motor test bench; integrated mechanical/electrical components and coordinated engineering/manufacturing requirements.

Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Capstone
Aerospace Engineering Team Member - In Progress | 2026-Present
- Contribute wing-model designs and project planning. Planned work includes CFD/thermal analysis, fabrication/testing, and fuel/battery-system integration; team flight target spring 2027.

SJSU Robotics | Mechanical Designer | Aug 2024-Present
- Designed a transmitter mast with camera/sensor mounts; used stress analysis to reinforce high-load regions and select lighter materials elsewhere.

FIRST Robotics Team 3925 | Electrical Lead - High School Robotics | 2020-2022
- Led circuitry design, battery testing, and circuit repairs for the team's robot; developed programming knowledge alongside electrical work.

BROADSWORD | Senior Design Rocket Concept | Mechanical & Programming Subgroup Lead | Aug-Dec 2023
- Designed landing legs in Inventor and coded stage liftoff/landing trajectories; coordinated mechanical/programming tasks for a student rocket concept.

SELECTED COURSEWORK

- AE 160 / AE 162 | Tested objects in an open wind tunnel; analyzed lift, drag, pitch, coefficients, and vortices; studied Navier-Stokes equations in aerodynamics coursework.
- ENGR 10 / AE 20 | Completed wind-tunnel blade CAD, aerodynamic/structural design, and power-efficiency studies; modeled a V6 piston engine in Inventor with paper schematics.
- AE 157 / AE 112 | Used MATLAB/Simulink for controls; tested structural/building designs and compared calculations in labs.

WORK EXPERIENCE

PayPal Stadium | Guest Representative | 2026 Soccer Season
- Worked San Jose Earthquakes and Bay FC women's team matches.

The Tech Interactive | Educator I | 2024-2025
- Taught STEM labs/design challenges and explained technical concepts to diverse learners.

Self-Employed | Private Swim Instructor | 2023
- Scheduled clients and tailored swim lessons to individual skills and goals.

YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021-2024
- Coached swimmers, led practices, and adapted instruction for children with disabilities.

LEADERSHIP & ATHLETICS

SJSU Division I Club Ultimate | 2022-Present: coordinate gym training and conditioning. SJSU Club Swimming | Spring 2026: competed at a Cal-hosted meet.